

Initiating Search

November 10, 2025, 4:25 PM

• Search:

CAS Lexicon Search:

Schiemann reaction - Preferred Concept

Search Tasks

Task	Result Type	View
Returned Reference Results from CAS Lexicon (64)	 References	View Results
Exported: Retrieved Related Reaction Results + Filters (83)	 Reactions	View Results
Filtered By:		
Yield:	90-100%, 80-89%, 70-79%	
Document Type:	Journal	

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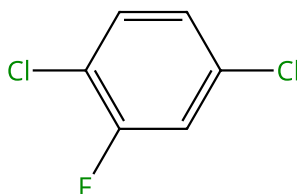
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 Reactions (50)[View in CAS SciFinder](#)

Scheme 1 (1 Reaction)

Steps: 1 Yield: 97%

Multi-component
structure image
available in CAS
SciFinder

 Suppliers (72) Suppliers (12)

31-190-CAS-8915085

Steps: 1 Yield: 97%

Continuous flow reactor for Balz-Schiemann reaction: a new procedure for the preparation of aromatic fluorides

By: Yu, Zhi-qun; et al

Tetrahedron Letters (2013), 54(10), 1261-1263.

1.1 Solvents: 1,2-Dichlorobenzene; 1 min, 200 °C; cooled

1.2 Reagents: Sodium hydroxide

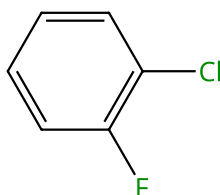
Solvents: Water; cooled

Experimental Protocols

Scheme 2 (1 Reaction)

Steps: 1 Yield: 97%

Multi-component
structure image
available in CAS
SciFinder

 Suppliers (49) Suppliers (16)

31-190-CAS-11660444

Steps: 1 Yield: 97%

Continuous flow reactor for Balz-Schiemann reaction: a new procedure for the preparation of aromatic fluorides

By: Yu, Zhi-qun; et al

Tetrahedron Letters (2013), 54(10), 1261-1263.

1.1 Solvents: 1,2-Dichlorobenzene; 1 min, 180 °C; cooled

1.2 Reagents: Sodium hydroxide

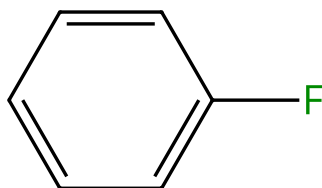
Solvents: Water; cooled

Experimental Protocols

Scheme 3 (6 Reactions)

Steps: 1 Yield: 79-97%

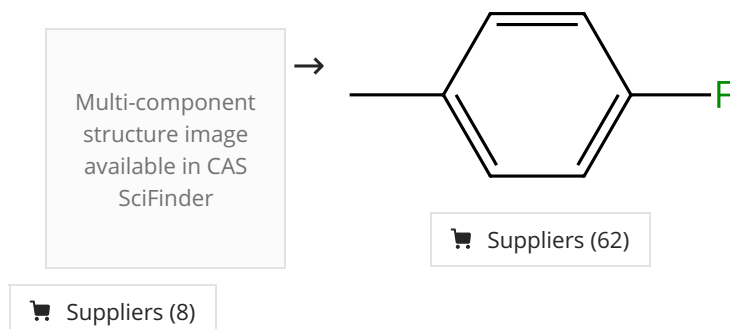
Multi-component
structure image
available in CAS
SciFinder

 Suppliers (88) Suppliers (25)

31-190-CAS-23914975 Steps: 1 Yield: 97% 1.1 Solvents: Chlorobenzene; 16 h, 60 °C 1.2 Reagents: Water Experimental Protocols	Revisiting the Balz-Schiemann Reaction of Aryldiazonium Tetrafluoroborate in Different Solvents under Catalyst- and Additive-Free Conditions By: Yang, Lian; et al ACS Omega (2021), 6(33), 21595-21603.
31-190-CAS-7119008 Steps: 1 Yield: 96% 1.1 1 min, 125 °C; cooled 1.2 Reagents: Sodium hydroxide Solvents: Water; cooled Experimental Protocols	Continuous flow reactor for Balz-Schiemann reaction: a new procedure for the preparation of aromatic fluorides By: Yu, Zhi-qun; et al Tetrahedron Letters (2013), 54(10), 1261-1263.
31-190-CAS-19049831 Steps: 1 Yield: 85% 1.1 Reagents: 1-Fluoro-1,3-dihydro-3,3-dimethyl-1,2-benziodoxole Catalysts: Boron trifluoride etherate Solvents: (Trifluoromethyl)benzene; 36 h, 35 °C Experimental Protocols	Hypervalent Iodine(III) Catalyzed Balz-Schiemann Fluorination under Mild Conditions By: Xing, Bo; et al Angewandte Chemie, International Edition (2018), 57(31), 9896-9900.
31-190-CAS-19049829 Steps: 1 Yield: 84% 1.1 Reagents: 1-(Difluorido)-4-methylbenzene Catalysts: Boron trifluoride etherate Solvents: Chlorobenzene; 36 h, 40 °C Experimental Protocols	Hypervalent Iodine(III) Catalyzed Balz-Schiemann Fluorination under Mild Conditions By: Xing, Bo; et al Angewandte Chemie, International Edition (2018), 57(31), 9896-9900.
31-190-CAS-19049828 Steps: 1 Yield: 81% 1.1 Reagents: 1-(Difluorido)-4-methylbenzene Catalysts: Boron trifluoride etherate Solvents: (Trifluoromethyl)benzene; 36 h, 35 °C Experimental Protocols	Hypervalent Iodine(III) Catalyzed Balz-Schiemann Fluorination under Mild Conditions By: Xing, Bo; et al Angewandte Chemie, International Edition (2018), 57(31), 9896-9900.
31-190-CAS-19049830 Steps: 1 Yield: 79% 1.1 Reagents: 3H-1,2-Benziodoxol-3-one, 1-fluoro- Catalysts: Boron trifluoride etherate Solvents: (Trifluoromethyl)benzene; 36 h, 35 °C Experimental Protocols	Hypervalent Iodine(III) Catalyzed Balz-Schiemann Fluorination under Mild Conditions By: Xing, Bo; et al Angewandte Chemie, International Edition (2018), 57(31), 9896-9900.

Scheme 4 (2 Reactions)

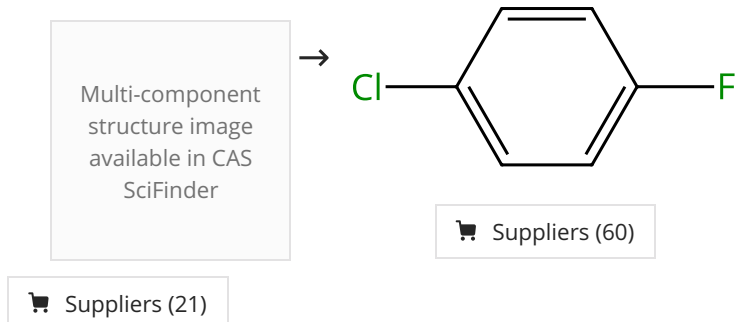
Steps: 1 Yield: 83-96%



31-190-CAS-6453943	Steps: 1 Yield: 96%	Continuous flow reactor for Balz-Schiemann reaction: a new procedure for the preparation of aromatic fluorides
1.1 1 min, 125 °C		By: Yu, Zhi-qun; et al
1.2 Reagents: Sodium hydroxide Solvents: Water; cooled		Tetrahedron Letters (2013), 54(10), 1261-1263.
Experimental Protocols		
31-190-CAS-19049832	Steps: 1 Yield: 83%	Hypervalent Iodine(III) Catalyzed Balz-Schiemann Fluorination under Mild Conditions
1.1 Reagents: 1-(Difluoriodo)-4-methylbenzene Catalysts: Boron trifluoride etherate Solvents: Chlorobenzene; 36 h, 50 °C		By: Xing, Bo; et al
Experimental Protocols		Angewandte Chemie, International Edition (2018), 57(31), 9896-9900.

Scheme 5 (1 Reaction)

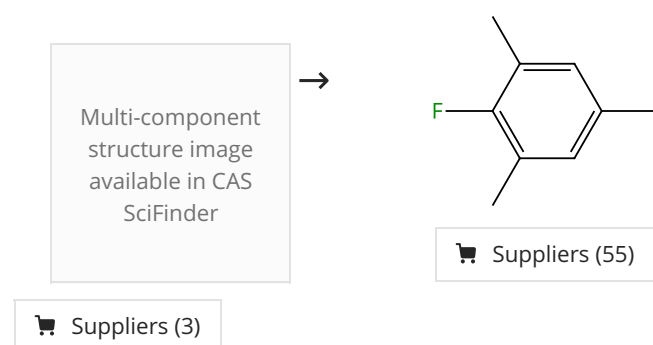
Steps: 1 Yield: 96%



31-190-CAS-6740431	Steps: 1 Yield: 96%	Continuous flow reactor for Balz-Schiemann reaction: a new procedure for the preparation of aromatic fluorides
1.1 1 min, 180 °C; cooled		By: Yu, Zhi-qun; et al
1.2 Reagents: Sodium hydroxide Solvents: Water; cooled		Tetrahedron Letters (2013), 54(10), 1261-1263.
Experimental Protocols		

Scheme 6 (3 Reactions)

Steps: 1 Yield: 94-95%

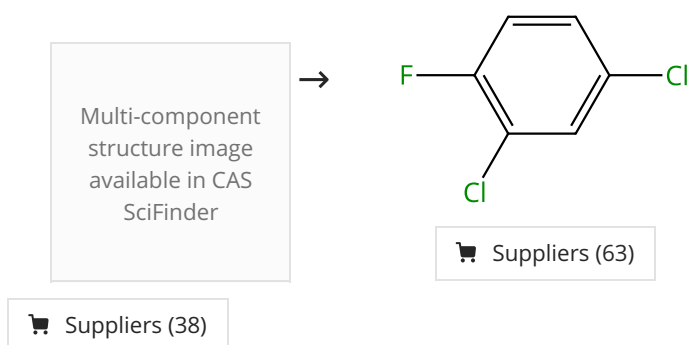


31-190-CAS-19049838	Steps: 1 Yield: 95%	Hypervalent Iodine(III) Catalyzed Balz-Schiemann Fluorination under Mild Conditions
1.1 Reagents: 1-(Difluoriodo)-4-methylbenzene Catalysts: Boron trifluoride etherate Solvents: Chlorobenzene; 36 h, 30 °C		By: Xing, Bo; et al
Experimental Protocols		Angewandte Chemie, International Edition (2018), 57(31), 9896-9900.

31-190-CAS-19049839 Steps: 1 Yield: 95% 1.1 Reagents: 3 <i>H</i> -1,2-Benziodoxol-3-one, 1-fluoro- Catalysts: Boron trifluoride etherate Solvents: Chlorobenzene; 36 h, 30 °C Experimental Protocols	Hypervalent Iodine(III) Catalyzed Balz-Schiemann Fluorination under Mild Conditions By: Xing, Bo; et al Angewandte Chemie, International Edition (2018), 57(31), 9896-9900.
31-190-CAS-19049840 Steps: 1 Yield: 94% 1.1 Reagents: 1-Fluoro-1,3-dihydro-3,3-dimethyl-1,2-benziodoxole Catalysts: Boron trifluoride etherate Solvents: Chlorobenzene; 36 h, 30 °C Experimental Protocols	Hypervalent Iodine(III) Catalyzed Balz-Schiemann Fluorination under Mild Conditions By: Xing, Bo; et al Angewandte Chemie, International Edition (2018), 57(31), 9896-9900.

Scheme 7 (1 Reaction)

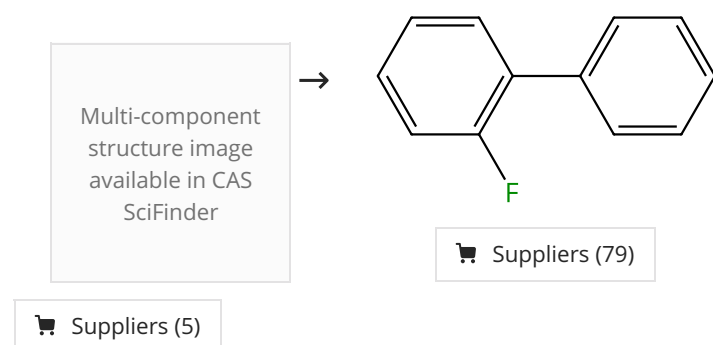
Steps: 1 Yield: 95%



31-190-CAS-348147 Steps: 1 Yield: 95% 1.1 Solvents: 1,2-Dichlorobenzene; 1 min, 190 °C; cooled 1.2 Reagents: Sodium hydroxide Solvents: Water; cooled Experimental Protocols	Continuous flow reactor for Balz-Schiemann reaction: a new procedure for the preparation of aromatic fluorides By: Yu, Zhi-qun; et al Tetrahedron Letters (2013), 54(10), 1261-1263.
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Scheme 8 (4 Reactions)

Steps: 1 Yield: 81-95%

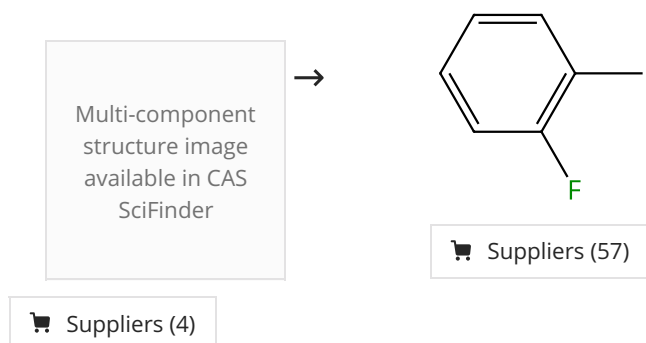


31-190-CAS-23916141 Steps: 1 Yield: 95% 1.1 Solvents: Chlorobenzene; 16 h, 60 °C 1.2 Reagents: Water Experimental Protocols	Revisiting the Balz-Schiemann Reaction of Aryldiazonium Tetrafluoroborate in Different Solvents under Catalyst- and Additive-Free Conditions By: Yang, Lian; et al ACS Omega (2021), 6(33), 21595-21603.
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31-190-CAS-19049853 Steps: 1 Yield: 90% 1.1 Reagents: 1-(Difluorido)-4-methylbenzene Catalysts: Boron trifluoride etherate Solvents: (Trifluoromethyl)benzene; 36 h, 25 °C Experimental Protocols	Hypervalent Iodine(III) Catalyzed Balz-Schiemann Fluorination under Mild Conditions By: Xing, Bo; et al Angewandte Chemie, International Edition (2018), 57(31), 9896-9900.
31-190-CAS-19049855 Steps: 1 Yield: 85% 1.1 Reagents: 1-Fluoro-1,3-dihydro-3,3-dimethyl-1,2-benziodoxole Catalysts: Boron trifluoride etherate Solvents: (Trifluoromethyl)benzene; 36 h, 25 °C Experimental Protocols	Hypervalent Iodine(III) Catalyzed Balz-Schiemann Fluorination under Mild Conditions By: Xing, Bo; et al Angewandte Chemie, International Edition (2018), 57(31), 9896-9900.
31-190-CAS-19049854 Steps: 1 Yield: 81% 1.1 Reagents: 3H-1,2-Benziodoxol-3-one, 1-fluoro- Catalysts: Boron trifluoride etherate Solvents: (Trifluoromethyl)benzene; 36 h, 25 °C Experimental Protocols	Hypervalent Iodine(III) Catalyzed Balz-Schiemann Fluorination under Mild Conditions By: Xing, Bo; et al Angewandte Chemie, International Edition (2018), 57(31), 9896-9900.

Scheme 9 (1 Reaction)

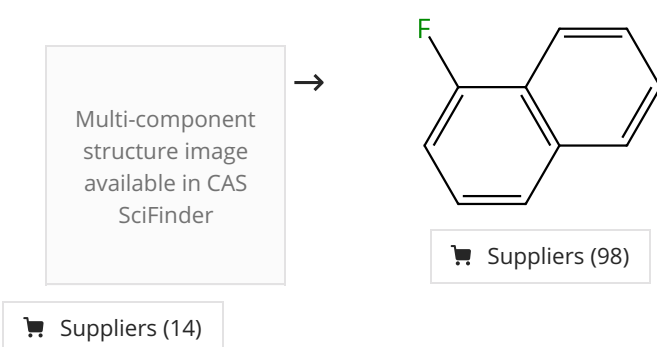
Steps: 1 Yield: 95%



31-190-CAS-11361111 Steps: 1 Yield: 95% 1.1 Solvents: 1,2-Dichlorobenzene; 1 min, 150 °C; cooled 1.2 Reagents: Sodium hydroxide Solvents: Water; cooled Experimental Protocols	Continuous flow reactor for Balz-Schiemann reaction: a new procedure for the preparation of aromatic fluorides By: Yu, Zhi-qun; et al Tetrahedron Letters (2013), 54(10), 1261-1263.
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Scheme 10 (4 Reactions)

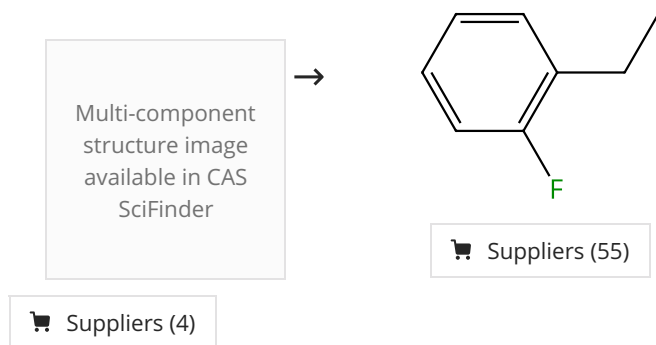
Steps: 1 Yield: 87-94%



31-190-CAS-19049849 Steps: 1 Yield: 94% 1.1 Reagents: 1-Fluoro-1,3-dihydro-3,3-dimethyl-1,2-benziodoxole Catalysts: Boron trifluoride etherate Solvents: (Trifluoromethyl)benzene; 36 h, 50 °C Experimental Protocols	Hypervalent Iodine(III) Catalyzed Balz-Schiemann Fluorination under Mild Conditions By: Xing, Bo; et al Angewandte Chemie, International Edition (2018), 57(31), 9896-9900.
31-190-CAS-19049848 Steps: 1 Yield: 93% 1.1 Reagents: 3H-1,2-Benziodoxol-3-one, 1-fluoro- Catalysts: Boron trifluoride etherate Solvents: (Trifluoromethyl)benzene; 36 h, 50 °C Experimental Protocols	Hypervalent Iodine(III) Catalyzed Balz-Schiemann Fluorination under Mild Conditions By: Xing, Bo; et al Angewandte Chemie, International Edition (2018), 57(31), 9896-9900.
31-190-CAS-19049847 Steps: 1 Yield: 90% 1.1 Reagents: 1-(Difluorido)-4-methylbenzene Catalysts: Boron trifluoride etherate Solvents: (Trifluoromethyl)benzene; 36 h, 50 °C Experimental Protocols	Hypervalent Iodine(III) Catalyzed Balz-Schiemann Fluorination under Mild Conditions By: Xing, Bo; et al Angewandte Chemie, International Edition (2018), 57(31), 9896-9900.
31-190-CAS-19049911 Steps: 1 Yield: 87% 1.1 Reagents: 1-Fluoro-1,3-dihydro-3,3-dimethyl-1,2-benziodoxole Catalysts: Boron trifluoride etherate Solvents: Chlorobenzene; 36 h, 50 °C; 50 °C → rt Experimental Protocols	Hypervalent Iodine(III) Catalyzed Balz-Schiemann Fluorination under Mild Conditions By: Xing, Bo; et al Angewandte Chemie, International Edition (2018), 57(31), 9896-9900.

Scheme 11 (1 Reaction)

Steps: 1 Yield: 92%

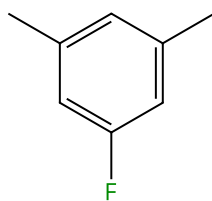


31-190-CAS-137998 Steps: 1 Yield: 92% 1.1 1 min, 100 °C; cooled 1.2 Reagents: Sodium hydroxide Solvents: Water; cooled Experimental Protocols	Continuous flow reactor for Balz-Schiemann reaction: a new procedure for the preparation of aromatic fluorides By: Yu, Zhi-qun; et al Tetrahedron Letters (2013), 54(10), 1261-1263.
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Scheme 12 (1 Reaction)

Steps: 1 Yield: 92%

Multi-component
structure image
available in CAS
SciFinder



Suppliers (62)

Supplier (1)

31-190-CAS-13459801

Steps: 1 Yield: 92%

Continuous flow reactor for Balz-Schiemann reaction: a new procedure for the preparation of aromatic fluorides

By: Yu, Zhi-qun; et al

Tetrahedron Letters (2013), 54(10), 1261-1263.

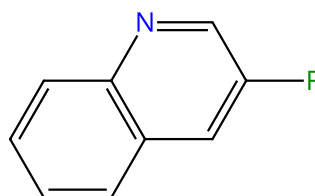
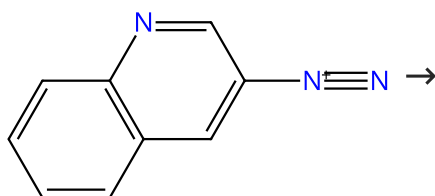
1.1 Solvents: Chlorobenzene; 1 min, 110 °C; cooled

1.2 Reagents: Sodium hydroxide
Solvents: Water; cooled

Experimental Protocols

Scheme 13 (1 Reaction)

Steps: 1 Yield: 91%



Suppliers (60)

31-190-CAS-15182663

Steps: 1 Yield: 91%

Preparation of heterocyclic fluorine compounds by the Schiemann reaction. II. Monofluoroquinolines

By: Roe, Arthur; et al

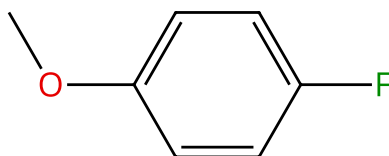
Journal of the American Chemical Society (1949), 71, 1785-6.

1.1 Solvents: Toluene

Scheme 14 (1 Reaction)

Steps: 1 Yield: 91%

Multi-component
structure image
available in CAS
SciFinder



Suppliers (75)

Suppliers (54)

31-190-CAS-10722389

Steps: 1 Yield: 91%

Continuous flow reactor for Balz-Schiemann reaction: a new procedure for the preparation of aromatic fluorides

By: Yu, Zhi-qun; et al

Tetrahedron Letters (2013), 54(10), 1261-1263.

1.1 1 min, 130 °C; cooled

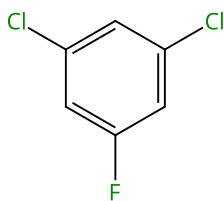
1.2 Reagents: Sodium hydroxide
Solvents: Water; cooled

Experimental Protocols

Scheme 15 (1 Reaction)

Steps: 1 Yield: 90%

Multi-component
structure image
available in CAS
SciFinder



Suppliers (64)

Suppliers (37)

31-190-CAS-2200361

Steps: 1 Yield: 90%

Continuous flow reactor for Balz-Schiemann reaction: a new procedure for the preparation of aromatic fluorides

By: Yu, Zhi-qun; et al

Tetrahedron Letters (2013), 54(10), 1261-1263.

1.1 Solvents: 1,2-Dichlorobenzene; 1 min, 160 °C; cooled

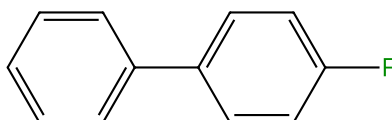
1.2 Reagents: Sodium hydroxide
Solvents: Water; cooled

Experimental Protocols

Scheme 16 (2 Reactions)

Steps: 1 Yield: 80-90%

Multi-component
structure image
available in CAS
SciFinder



Suppliers (76)

Suppliers (2)

31-190-CAS-23912033

Steps: 1 Yield: 90%

Revisiting the Balz-Schiemann Reaction of Aryldiazonium Tetrafluoroborate in Different Solvents under Catalyst- and Additive-Free Conditions

By: Yang, Lian; et al

ACS Omega (2021), 6(33), 21595-21603.

1.1 Solvents: Chlorobenzene; 16 h, 70 °C

1.2 Reagents: Water

Experimental Protocols

31-190-CAS-19049856

Steps: 1 Yield: 80%

Hypervalent Iodine(III) Catalyzed Balz-Schiemann Fluorination under Mild Conditions

By: Xing, Bo; et al

Angewandte Chemie, International Edition (2018), 57(31), 9896-9900.

1.1 Reagents: 1-(Difluorido)-4-methylbenzene

Catalysts: Boron trifluoride etherate

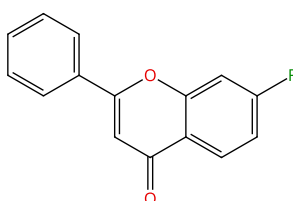
Solvents: Chlorobenzene; 36 h, 60 °C

Experimental Protocols

Scheme 17 (1 Reaction)

Steps: 1 Yield: 87%

Multi-component
structure image
available in CAS
SciFinder

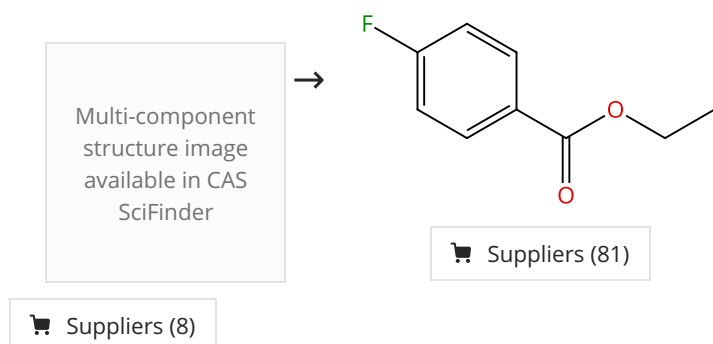


Suppliers (8)

31-190-CAS-23914871 Steps: 1 Yield: 87% 1.1 Solvents: Chlorobenzene; 96 h, 80 °C 1.2 Reagents: Water Experimental Protocols	Revisiting the Balz-Schiemann Reaction of Aryldiazonium Tetrafluoroborate in Different Solvents under Catalyst- and Additive-Free Conditions By: Yang, Lian; et al ACS Omega (2021), 6(33), 21595-21603.
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Scheme 18 (2 Reactions)

Steps: 1 Yield: 82-87%

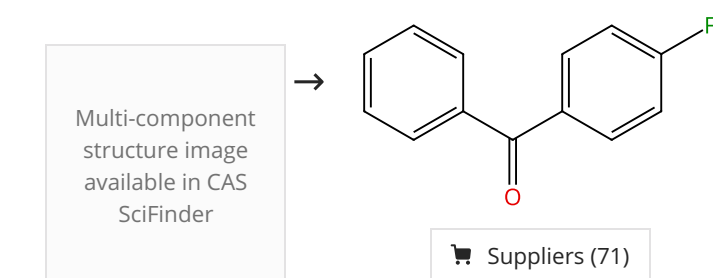


31-190-CAS-19049870 Steps: 1 Yield: 87% 1.1 Reagents: 1-(Difluorido)-4-methylbenzene Catalysts: Boron trifluoride etherate Solvents: Chlorobenzene; 36 h, 55 °C Experimental Protocols	Hypervalent Iodine(III) Catalyzed Balz-Schiemann Fluorination under Mild Conditions By: Xing, Bo; et al Angewandte Chemie, International Edition (2018), 57(31), 9896-9900.
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31-190-CAS-19049871 Steps: 1 Yield: 82% 1.1 Reagents: 3H-1,2-Benziodoxol-3-one, 1-fluoro- Catalysts: Boron trifluoride etherate Solvents: Chlorobenzene; 36 h, 55 °C Experimental Protocols	Hypervalent Iodine(III) Catalyzed Balz-Schiemann Fluorination under Mild Conditions By: Xing, Bo; et al Angewandte Chemie, International Edition (2018), 57(31), 9896-9900.
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Scheme 19 (1 Reaction)

Steps: 1 Yield: 85%

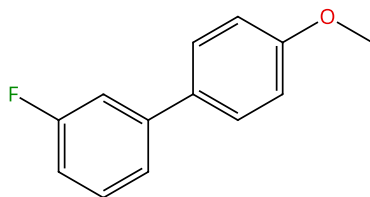


31-190-CAS-23914124 Steps: 1 Yield: 85% 1.1 24 h, 70 °C 1.2 Reagents: Water Experimental Protocols	Revisiting the Balz-Schiemann Reaction of Aryldiazonium Tetrafluoroborate in Different Solvents under Catalyst- and Additive-Free Conditions By: Yang, Lian; et al ACS Omega (2021), 6(33), 21595-21603.
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Scheme 20 (1 Reaction)

Steps: 1 Yield: 85%

Multi-component
structure image
available in CAS
SciFinder



Suppliers (29)

31-190-CAS-23912408

Steps: 1 Yield: 85%

Revisiting the Balz-Schiemann Reaction of Aryldiazonium
Tetrafluoroborate in Different Solvents under Catalyst- and
Additive-Free Conditions

1.1 Solvents: Chlorobenzene; 24 h, rt

1.2 Reagents: Water

Experimental Protocols

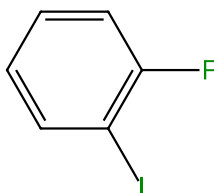
By: Yang, Lian; et al

ACS Omega (2021), 6(33), 21595-21603.

Scheme 21 (1 Reaction)

Steps: 1 Yield: 85%

Multi-component
structure image
available in CAS
SciFinder



Suppliers (86)

Suppliers (2)

31-190-CAS-19049858

Steps: 1 Yield: 85%

Hypervalent Iodine(III) Catalyzed Balz-Schiemann Fluorination
under Mild Conditions

1.1 Reagents: 1-(Difluorido)-4-methylbenzene

Catalysts: Boron trifluoride etherate

Solvents: (Trifluoromethyl)benzene; 36 h, 45 °C

Experimental Protocols

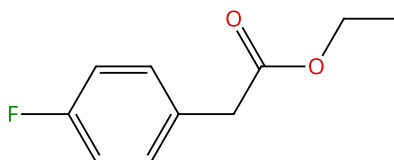
By: Xing, Bo; et al

Angewandte Chemie, International Edition (2018), 57(31),
9896-9900.

Scheme 22 (1 Reaction)

Steps: 1 Yield: 85%

Multi-component
structure image
available in CAS
SciFinder



Suppliers (64)

31-190-CAS-23912611

Steps: 1 Yield: 85%

Revisiting the Balz-Schiemann Reaction of Aryldiazonium
Tetrafluoroborate in Different Solvents under Catalyst- and
Additive-Free Conditions

1.1 Solvents: Chlorobenzene; 16 h, 60 °C

1.2 Reagents: Water

Experimental Protocols

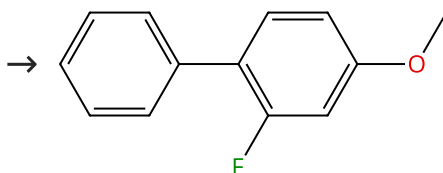
By: Yang, Lian; et al

ACS Omega (2021), 6(33), 21595-21603.

Scheme 23 (1 Reaction)

Steps: 1 Yield: 84%

Multi-component
structure image
available in CAS
SciFinder



Suppliers (19)

31-190-CAS-23911415

Steps: 1 Yield: 84%

Revisiting the Balz-Schiemann Reaction of Aryldiazonium
Tetrafluoroborate in Different Solvents under Catalyst- and
Additive-Free Conditions

1.1 Solvents: Chlorobenzene; 3 h, 60 °C

1.2 Reagents: Water

Experimental Protocols

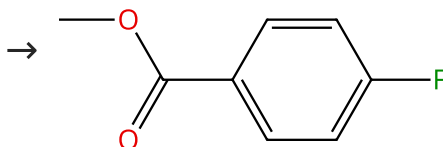
By: Yang, Lian; et al

ACS Omega (2021), 6(33), 21595-21603.

Scheme 24 (1 Reaction)

Steps: 1 Yield: 83%

Multi-component
structure image
available in CAS
SciFinder



Suppliers (88)

Suppliers (6)

31-190-CAS-23912737

Steps: 1 Yield: 83%

Revisiting the Balz-Schiemann Reaction of Aryldiazonium
Tetrafluoroborate in Different Solvents under Catalyst- and
Additive-Free Conditions

1.1 Solvents: Chlorobenzene; 16 h, 70 °C

1.2 Reagents: Water

Experimental Protocols

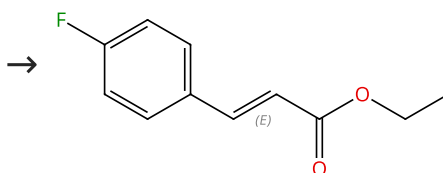
By: Yang, Lian; et al

ACS Omega (2021), 6(33), 21595-21603.

Scheme 25 (1 Reaction)

Steps: 1 Yield: 83%

Multi-component
structure image
available in CAS
SciFinder



Double bond geometry shown

Suppliers (47)

31-190-CAS-23912646

Steps: 1 Yield: 83%

Revisiting the Balz-Schiemann Reaction of Aryldiazonium
Tetrafluoroborate in Different Solvents under Catalyst- and
Additive-Free Conditions

1.1 Solvents: Hexane; 58 h, 70 °C

1.2 Reagents: Water

Experimental Protocols

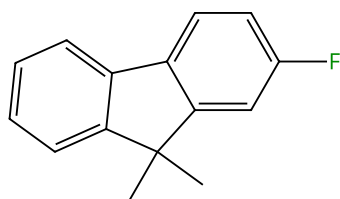
By: Yang, Lian; et al

ACS Omega (2021), 6(33), 21595-21603.

Scheme 26 (1 Reaction)

Steps: 1 Yield: 83%

Multi-component
structure image
available in CAS
SciFinder



Supplier (1)

31-190-CAS-23915064

Steps: 1 Yield: 83%

Revisiting the Balz-Schiemann Reaction of Aryldiazonium
Tetrafluoroborate in Different Solvents under Catalyst- and
Additive-Free Conditions

1.1 Solvents: Chlorobenzene; 48 h, 80 °C

1.2 Reagents: Water

Experimental Protocols

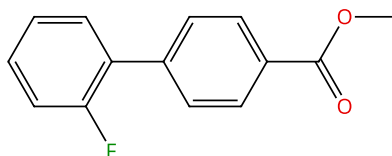
By: Yang, Lian; et al

ACS Omega (2021), 6(33), 21595-21603.

Scheme 27 (1 Reaction)

Steps: 1 Yield: 82%

Multi-component
structure image
available in CAS
SciFinder



Suppliers (29)

31-190-CAS-23914981

Steps: 1 Yield: 82%

Revisiting the Balz-Schiemann Reaction of Aryldiazonium
Tetrafluoroborate in Different Solvents under Catalyst- and
Additive-Free Conditions

1.1 Solvents: Hexane; 16 h, 60 °C

1.2 Reagents: Water

Experimental Protocols

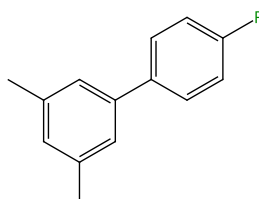
By: Yang, Lian; et al

ACS Omega (2021), 6(33), 21595-21603.

Scheme 28 (1 Reaction)

Steps: 1 Yield: 82%

Multi-component
structure image
available in CAS
SciFinder



Suppliers (2)

31-190-CAS-23915941

Steps: 1 Yield: 82%

Revisiting the Balz-Schiemann Reaction of Aryldiazonium
Tetrafluoroborate in Different Solvents under Catalyst- and
Additive-Free Conditions

1.1 Solvents: Hexane; 48 h, 80 °C

1.2 Reagents: Water

Experimental Protocols

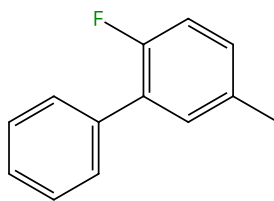
By: Yang, Lian; et al

ACS Omega (2021), 6(33), 21595-21603.

Scheme 29 (1 Reaction)

Steps: 1 Yield: 81%

Multi-component
structure image
available in CAS
SciFinder



Suppliers (22)

31-190-CAS-23915112

Steps: 1 Yield: 81%

Revisiting the Balz-Schiemann Reaction of Aryldiazonium Tetrafluoroborate in Different Solvents under Catalyst- and Additive-Free Conditions

1.1 Solvents: Chlorobenzene; 3 h, 60 °C

1.2 Reagents: Water

Experimental Protocols

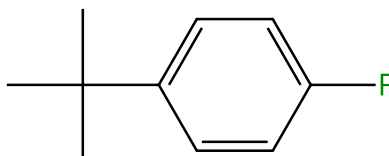
By: Yang, Lian; et al

ACS Omega (2021), 6(33), 21595-21603.

Scheme 30 (3 Reactions)

Steps: 1 Yield: 80-81%

Multi-component
structure image
available in CAS
SciFinder



Suppliers (39)

Suppliers (4)

31-190-CAS-19049835

Steps: 1 Yield: 81%

Hypervalent Iodine(III) Catalyzed Balz-Schiemann Fluorination under Mild Conditions

1.1 Reagents: 1-(Difluorido)-4-methylbenzene

Catalysts: Boron trifluoride etherate

Solvents: (Trifluoromethyl)benzene; 36 h, 40 °C

Experimental Protocols

By: Xing, Bo; et al

Angewandte Chemie, International Edition (2018), 57(31), 9896-9900.

31-190-CAS-19049837

Steps: 1 Yield: 81%

Hypervalent Iodine(III) Catalyzed Balz-Schiemann Fluorination under Mild Conditions

1.1 Reagents: 1-Fluoro-1,3-dihydro-3,3-dimethyl-1,2-benziodoxole

Catalysts: Boron trifluoride etherate

Solvents: (Trifluoromethyl)benzene; 36 h, 40 °C

Experimental Protocols

By: Xing, Bo; et al

Angewandte Chemie, International Edition (2018), 57(31), 9896-9900.

31-190-CAS-19049836

Steps: 1 Yield: 80%

Hypervalent Iodine(III) Catalyzed Balz-Schiemann Fluorination under Mild Conditions

1.1 Reagents: 3H-1,2-Benziodoxol-3-one, 1-fluoro-

Catalysts: Boron trifluoride etherate

Solvents: (Trifluoromethyl)benzene; 36 h, 40 °C

Experimental Protocols

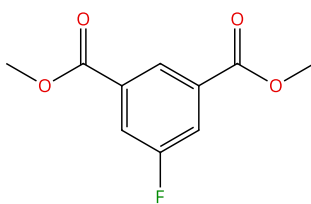
By: Xing, Bo; et al

Angewandte Chemie, International Edition (2018), 57(31), 9896-9900.

Scheme 31 (1 Reaction)

Steps: 1 Yield: 81%

Multi-component
structure image
available in CAS
SciFinder



Suppliers (51)

31-190-CAS-19049903

Steps: 1 Yield: 81%

Hypervalent Iodine(III) Catalyzed Balz-Schiemann Fluorination under Mild Conditions

By: Xing, Bo; et al

Angewandte Chemie, International Edition (2018), 57(31), 9896-9900.

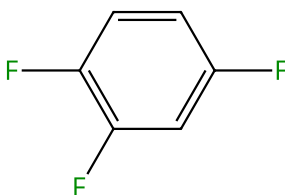
1.1 **Reagents:** 1-Fluoro-1,3-dihydro-3,3-dimethyl-1,2-benziodoxole
Catalysts: Boron trifluoride etherate
Solvents: Chlorobenzene; 36 h, 80 °C

Experimental Protocols

Scheme 32 (1 Reaction)

Steps: 1 Yield: 79%

Multi-component
structure image
available in CAS
SciFinder



Suppliers (71)

Suppliers (3)

31-190-CAS-4655115

Steps: 1 Yield: 79%

Continuous flow reactor for Balz-Schiemann reaction: a new procedure for the preparation of aromatic fluorides

By: Yu, Zhi-qun; et al

Tetrahedron Letters (2013), 54(10), 1261-1263.

1.1 **Solvents:** 1,2-Dichlorobenzene; 1 min, 200 °C; cooled

1.2 **Reagents:** Sodium hydroxide
Solvents: Water; cooled

Experimental Protocols